

Curriculum Vitae

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EMPLOYMENT HISTORY

- 2021– Professor, Department of Automation and Industrial Informatics,
National University of Science and Technology "Politehnica" of Bucharest
Head of Artificial Intelligence for SENSing, Low-Power Networking and ICS SEcurity ([AISENSE](#))
Laboratory - PRECIS Research Institute
- 2020– Founder and Research Scientist, [Applied Research Solutions](#)
- 2021–2024 ICT Consultant, TÜV Austria Romania
- 2017–2021 Associate Professor, Department of Automation and Industrial Informatics,
University "Politehnica" of Bucharest
Habilitation certificate (Dr.-Ing. habil.) in Automation and Industrial Informatics
Thesis: *Information Processing in Distributed Sensor Systems*
- 2019 University Project Assistant (with Doctorate), Institute of Technical Informatics,
Technical University of Graz, Austria. LEAD Project "Dependable Things in Adverse Environments"
- 2015–2016 Fulbright Visiting Scholar, EECS Department, School of Engineering,
University of California, Merced, USA
Topic: *Pervasive Building Energy Monitoring and Control over Reliable Wireless Sensor Networks*
Faculty associate: Prof. Alberto Cerpa
- 2014–2015 Postdoctoral Researcher, Department of Automation and Industrial Informatics,
University "Politehnica" of Bucharest
Topic: *Aggregation and Data Fusion Strategies in Multi-sensory Intelligent Environments*, Project:
"Development of Human Resources in Doctoral and Postdoctoral Research: Driving Force for a Knowledge-
based Society" - KNOWLEDGE, POSDRU/159/1.5/S/134398. Adviser: Prof. Dan Popescu
- 2009–2017 Teaching Assistant and Assistant Professor (2013+), Department of Automation and Industrial Informatics, University "Politehnica" of Bucharest

EDUCATION

- 2012 Ph. D., University "Politehnica" of Bucharest, Faculty of Automatic Control and Computers, Systems Engineering (adviser: Prof. Dr. Ing. Valentin Sgârțiu), Public defense date: 19.10.2012
Thesis: *Improving Life and Work with Reliable Wireless Sensor Networks*. Grade: *Magna cum laudae*
- 2011 PhD Research Internship, Distributed Embedded Systems Group, Institute of Computer Engineering - ITI, University of Lübeck, Germany, *Automating data collection and analysis from laboratory mice used in medical research through light-weight camera-enabled sensor networks and distributed computing (MiceNet)*, Adviser: Prof. Kay Römer
- 2011 M. Sc., University "Politehnica" of Bucharest, Faculty of Automatic Control and Computers, Automatic Control and Industrial Informatics
- 2009 ERASMUS Student, LaS3 Laboratory, University of Applied Sciences Regensburg, Germany, Diploma thesis: *Research on the influence of caching on task migration in multicore processors used in embedded automotive real-time systems*, Advisers: Prof. Jürgen Mottok (FH R), Prof. Dumitru Popescu (UPB)
- 2009 Dipl.-Eng. (5 year Engineering Studies, M.Sc. equivalent), University "Politehnica" of Bucharest, Faculty of Automatic Control and Computers, *top 2% – ranked 4th out of 259 graduates*

AWARDS

- 2024 [Distinguished Lecturer of the IEEE Instrumentation and Measurement Society](#) 2025-2027, Topic: "Big Measurement Data".
- 2018 Best Paper Award, *Wireless Sensor Network Architecture based on Fog Computing*, CODIT 2018, Thessaloniki, Greece.
- 2013 1st Prize, NIDays 2013 Graphical System Design Conference, Bucharest, October 2013.
Competitive Travel Grant, 2nd COST IntelliCIS Training School – Simulation-based Design of Complex Infrastructure Systems, RWTH Aachen, Germany.
- 2012 Competitive Travel Grant, Advanced School on ICT for Future Energy Systems (ICT4FES'12), Trento.
Best Paper Award, SENSORCOMM 2012, Rome, Italy.
Student Travel Grant, 9th European Wireless Sensor Networks Conference – EWSN 2012, Trento, Italy.
- 2011 "Are Hjørungnes" Student Travel Grant, IEEE Distributed Computing in Sensor Systems, Barcelona.

RESEARCH PROJECTS - PAST 5 YEARS

- 2024–2027 Researcher, *Novel current control for climate neutral energy infrastructure*, Horizon Europe Call CL5-2024-D2-01, GA 101192615. [NOVETROL](#)
- 2024–2026 Researcher, *Information Fusion of Multi-vector Real-Time Data Streams for Energy Management in Emerging Power Grids*, ERA-NET Enerdigite Call 2023. [INFUSE](#)
- 2024–2026 Researcher, *FIWARE Driven Energy Communities for the Future*, Clean Energy Transition (CET) Partnership, Joint Call 2022. [ECOMM4FUTURE](#)
- 2024 **Project partner responsible**, *Resource Efficient Manufacturing Model Development and Implementation*, GreenSME Open Call 1. Beneficiary: Asti Automation. [REFFICIENT](#)
- 2023 **Project responsible**, *Cognitive Robotic Cell through Embedded ML Vision*, StairwAI Open Call 3. Beneficiary: Asti Automation. [COGNICELL](#)
- 2022–2023 Researcher, *Bridging Carbon Neutral LNG with the Energy Market – A Novel Ontochain Application*, NGI ONTOCHAIN Open Call 3. Beneficiary: Applied Research Solutions. [OTCnLNG](#)
- 2021–2024 **Project partner responsible**, *rEsilient and seLf-healed EleCTRical pOwer Nanogrid*, Call: H2020-SU-DS-2020, Topic: U-DS04-2018-2020, Start Date: 01/10/2021, Duration: 36 months. Partner: TUV Austria Romania, \$257075. [ELECTRON](#)
- 2021 **Project director**, *PROSIM Cyber-Physical Platform for Application Development and Training in the Process Industries*, HUBCAP Call #2 Experiment, Subgrant no. HUBCAP-OC2.1-2020/1499086, Beneficiary: Asti Automation, \$65558. [PRO-CPS](#)
- 2021–2023 Researcher, *Extending the Measurement Concept for the Control of Emerging Power Systems – EMERGE*, UEFISCDI Exploratory Research Projects Call 2020, PN-III-P4-ID-PCE- 2020-2876. [EMERGE](#)
- 2020–2023 Researcher, *HALYomorpha halys IDentification: Innovative ICT tools for targeted monitoring and sustainable management of the brown marmorated stink bug and other pests – HALY.ID*, HORIZON 2020, ERA NET COFUND, CT-AGRI-FOOD, 2019 Joint Call, ID 40606. [HALY.ID](#)

PUBLICATIONS

Over 150 publications: books/book chapters, articles in journals and conference proceedings, in the fields of wireless sensor networks, information processing and intelligent measurements. *See on-line researcher profiles for full list.*

- P10. Craioveanu, G. M., **Stamatescu, G.**, Saukh, O., *Look, Reason, Defuse: Bridging Perception and Domain Knowledge for Real-World Unexploded Ordnance Identification*. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 9113-9124, 2026.
- P9. Pavel M.-D., **Stamatescu G.**, Chodnicki M., Amza C.G., *LLM-Enhanced Control of a Mobile Robotic Platform for Smart Industry*. Appl. Sci. 2026, 16, 1680.

- P8. Craioveanu M., **Stamatescu G.**, Popescu D., *Ensemble Strategy With Multi-Step Hard Sample Mining for Improved UXO Localisation and Classification*, IEEE Access, vol. 13, pp. 123546-123558, 2025.
- P7. Gresoi S., **Stamatescu G.**, Fagarasan I., *Advanced Methodology for Fraud Detection in Energy Using Machine Learning Algorithms*, Applied Sciences 2025, 25, 3361.
- P6. Cretu G., Stamatescu I., **Stamatescu G.**, *Modeling and prediction of occupancy in buildings based on sensor data using deep learning methods*, IEEE Access, vol. 12, pp. 102994-103003, 2024.
- P5. M.D. Pavel, **Stamatescu G.**, *Flexible Manufacturing System for Enhanced Industry 4.0 and Industry 5.0 Applications*, 20th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT), Abu Dhabi, United Arab Emirates, pp. 483-490, 2024.
- P4. Cretu G., Stamatescu I., **Stamatescu G.**, *Evaluation of Deep Learning and Machine Learning Algorithms for Building Occupancy Classification on Open Datasets*, 31st Mediterranean Conference on Control and Automation (MED 2023), Limassol, June 2023.
- P3. **Stamatescu G.**, Ciornei I., Plamanescu R., Albu M., *Detection of Anomalies in Power Profiles using Data Analytics*, 12th IEEE International Workshop on Applied Measurements for Power Systems (AMPS 2022), Cagliari, September 2022.
- P2. Fagaras R., Nichiforov C., Stamatescu I., **Stamatescu G.**, *Evaluation of Compressed Residential Energy Forecasting Models*, 2021 IEEE International Conference on Systems, Man, and Cybernetics (SMC 2021), Melbourne, October 2021.
- P1. **Stamatescu G.**, Chitu C., *Privacy-Preserving Sensing and Two-Stage Building Occupancy Prediction Using Random Forest Learning*, Journal of Sensors, vol. 2021, Article ID 8000595, 2021.

PROFESSIONAL ACTIVITIES

Service selection

- Lead Guest Editor: Special Issue on Intelligent Sensors and Advanced Computing: Developments in the Era of Industry 4.0: 2nd Edition ([IDAACS2025](#)), Special Issue on Complex Data Processing Systems and Computing Algorithms: New Concepts and Applications ([DataPro](#)); Special Issue on Convergence of Intelligent Data Acquisition and Advanced Computing Systems ([IDAACS2019](#)), *Sensors (MDPI)*; Special Issue on Sensing and Data-Driven Control for Smart Building and Smart City Systems ([SBSCS](#)), *Journal of Sensors*. Guest Editor: Special Issue on Advanced Topics in Systems Safety and Security ([IWSSS2019](#)), *Information*; Special Issue on Energy Management for Smart Buildings ([EM SB](#)), *Energies*, Special Issue on Privacy-Preserving and System Security Control Based on Machine Learning ([PPSSCML](#)), *Applied Sciences*. Past associate editor: Journal of Sensors, IEEE ACCESS, Sensors.
- General co-chair, *9th IEEE International Conference on Intelligent Data Acquisition and Advanced Computing Systems: Technology and Applications IDAACS 2017*; Organizing Committee Co-Chair, *International Workshop on Systems Safety & Security (IWSSS)*, 2013-2025; Special sessions: Advanced Methods and Applications for Increasing Control Systems Reliability, Safety and Security - IEEE SMC 2016, Intelligent Wireless Sensor Networks for Monitoring, Diagnosis and Control - IEEE CODIT 2017 and CODIT 2018, Control in Smart Cities - MED 2018, Intelligent data processing from sensors in control and decision support systems - MED 2021, Special Stream in Smart Buildings and Smart Cities - IDAACS 2019/2021/2023, Workshop on Digitalisation of Manufacturing and Collaborative Robotics for Industry 5.0 - ETFA 2023.
- Reviewer for several relevant top journals such as IEEE Sensors, Elsevier Engineering Applications of Artificial Intelligence, Computers in Industry, Applied Energy, Sustainable Cities and Society. Member of multiple TPCs for international indexed conferences, most recently IEEE SECON, IEEE MASS, IEEE DCOSS-IOT.
- External expert/evaluator for the Ministry of Scientific Research and Innovation, Ministry of Energy, UEFIS-CDI, Malta Council for Science and Technology, Foundation for Polish Science, NSERC, RIF Cyprus, Science Fund of the Republic of Serbia (SFRS), EU-COST, DG-REGIO, EURAMET, European Commission Horizon Europe Program, EU TAIEX Expert in the fields of Artificial Intelligence and Cybersecurity.

- Certified trainer for the Romanian National Institute of Administration (INA) – ITC Domain: Use of AI Tools and Cybersecurity training programs.

Memberships

- 2007– Senior Member, Institute of Electrical and Electronics Engineers (IEEE) – [IEEE Romania Section Chair](#) (Secretary 2021 – 2025). Past-Chair of [IEEE Romania Robotics and Automation Society Chapter](#). Chair of IEEE Romania Section Young Professionals (YP) Affinity Group 2017-2019. Member of IEEE RAS [Technical Committee on Smart Buildings](#). Member of IEEE IES [Technical Committee on Industrial Agents](#). Corresponding member of IEEE Region 8 [Membership Development Committee](#).
- 2016– Senior Member, Association for Computing Machinery (ACM); Special Interest Group on Energy Systems and Informatics ([SIGEnergy](#)).
- 2017– Member, IFAC TC3.3. Telematics: Control via Communication Networks.
- 2025– Member, CIGRE D2.64 Working Group on Application of AI in Cybersecurity Defence of Power Systems.

Invited Talks *selection*

- 2025 *What's next for AI-based Cybersecurity of Power Systems?*, European Energy-ISAC (EE-ISAC) 26th Anniversary Plenary, Brussels, Belgium, October 2025.
- 2024 *Complementary View on the NIS2 Directive and the Network Code on Cybersecurity: Technology-enabled Best Practices*, Critical Infrastructure Protection and Resilience Europe (CIPRE), Madrid, Spain, November 2024.
- 2024 *AI-Enhanced Cybersecurity for Energy: The ELECTRON Platform*, Big Data Week, Bucharest, Romania, October 2024.
- 2023 *European Regulatory and Policy Landscape for Cybersecurity of Energy Systems*, MED'23 Workshop on Cybersecurity for Electrical Power and Energy Systems (EPES), Limassol, Cyprus, June 2023.
- 2023 *Tutorial: Detection and Forecasting of Micro-Scale Variability in Electric Power Profiles*, 2023 IEEE International Instrumentation and Measurement Technology Conference (I2MTC 2023), Kuala Lumpur, Malaysia, May 2023. *with M. Albu*
- 2023 *Data Pre-processing and Automated Machine learning for Energy AI: An Overview*, 10th International Conference on Signal Processing and Integrated Networks (SPIN 2023), Amity University, India, March 2023.
- 2022 *Tutorial: Data Preprocessing Techniques for Multiscale Measurements Analytics*, 2022 IEEE International Instrumentation and Measurement Technology Conference (I2MTC 2022), Ottawa, Canada, May 2022. *with M. Albu*
- 2019 *Deep Impact Models for Local Characterisation of Energy Events*, Department of Automation – Tsinghua University, December 6th, Beijing, China.

DEPARTMENT ACTIVITIES AND INSTITUTIONAL DEVELOPMENT

Advising students for project work, student scientific research competitions, bachelor and master thesis. Research activities in the field of automation and industrial informatics with a focus on intelligent data processing, distributed sensing and the industrial internet of things. Preparing scientific conference proceedings and journal articles, project reports and giving scientific talks. Reporting and administrative work. Social and educational outreach, industry collaboration and technology transfer.

- 2014–2015 Supervising faculty, Institutional development project co-financed with european funds, POSDRU 161 /2.1/G/134386, *From school, let's prepare ourselves for an active life*, enabling student career counseling and practical experience through specialised internships. [SHIVA](#)
- 2010–2017 Co-founder and Coordinator, *National Instruments LabVIEW Academy Program*, Faculty of Automatic Control and Computers, UPB, First implementation of the NI academic program in Romania.
- 2013–2017 Co-founder and member of the team, *Phoenix Contact EduNet International Education Network*, Faculty of Automatic Control and Computers, UPB, First implementation of EduNet in Romania.

- 2020–2024 Member of the Council of the Faculty of Automatic Control and Computers: committee for research, innovation and relation with the business environment and committee for international relations.
- 2022–2026 Member of the [Worlddidac](#) Council: the global trade association for the educational resources industry.
- 2024–2028 Member of the Romanian National Council for the Attestation of University Degrees, Diplomas and Certificates, Committee 15 on Computers, Information Technology and Systems Engineering.
- 2024–2029 Member of the Council of the Department of Automation and Industrial Informatics, UPB.
- 2021–2022 [Engineer your Future!](#) STEM educational outreach project for high-school seniors. Small Grants for Alumni Programs – Embassy of the United States of America in Romania.
- 2020–2024 Project coordinator for Partner UPB, Erasmus+ Capacity Building Project *Master of Engineering in Internet of Things*, Coordinator: University of Siegen, Germany. [IoTRAIN](#).
- 2024–2026 Erasmus+ Cooperation Partnerships in Higher Education Project *Building Opportunities for SMEs Sustainable Entrepreneurship with Artificial Intelligence in Industry 5.0*, Partner: [Asti Automation](#), Romania. [BOOST-AI](#).
- 2025 Honorary Advisor, Romanian Ministry of Energy, Digitalization and Cybersecurity topics.
- 2026–2027 Erasmus+ Capacity building in the field of Vocational Education and Training (VET) Project *Collaborative Robots Vocational Training in Algeria and Tunisia*, Partner: [Asti Automation](#), Romania. [CobostVETMed](#).

TEACHING

- 2013– Regelungstechnik I/II (Control Engineering 1/2), 3rd/4th year B.Sc., Faculty of Engineering in Foreign Languages.
- 2020– Intelligent Measurement Systems, Master programme of Automation and Industrial Informatics, Faculty of Automatic Control and Computers
- 2023– Computerprogrammierung und Programmiersprachen I/II (Computer Programming and Programming Languages 1/2), 1st year B.Sc., Faculty of Entrepreneurship, Business Engineering and Management
- 2024– Industrial Control Systems Cybersecurity, Master programme of Automation and Industrial Informatics, Faculty of Automatic Control and Computers
- 2024– Artificial Intelligence, 4th year B.Sc., Faculty of Automatic Control and Computers and Faculty of Industrial Engineering and Robotics
- 2025– Forecasting and Decision Support Systems, Master programme of Advanced Analytics for Business, Faculty of Automatic Control and Computers

FOREIGN LANGUAGES

English (C2), Spanish (C2), German (C2), French (B2), Russian (A2).